

Assignment 3: No-Code Platforms for Single-Cell & Spatial Omics Analysis

Explore one platform, test it with a real dataset, and prepare a short presentation based on your findings.

Objective

- Understand how no-code or low-code tools can be used for single-cell or spatial omics analysis.
- Test one platform using a real or example dataset.
- Prepare slides that explain the platform, workflow, strengths, and limitations.

Assigned Platforms

Student	Platform	Website	Main Use
1	Loupe Browser	https://www.10xgenomics.com/support/software/loupe-browser/downloads	10x single-cell and Visium/spatial exploration
2	Galaxy Single Cell & Spatial Omics	https://singlecell.usegalaxy.eu/	Online no-code workflows for single-cell and some spatial analysis
3	scExplorer	https://apps.cienciavida.org/scexplorer/	Online scRNA-seq analysis and visualization
4	Nygen / ScarfWeb	https://www.nygen.io/products/scarfweb	AI-assisted browser-based single-cell analysis; check free account/demo access

What to Check

- Platform overview: purpose, developer/company, online or desktop, free/demo/paid status.
- Supported data types: scRNA-seq, spatial transcriptomics, Visium, multiomics, ATAC-seq, etc.
- Input formats: h5ad, h5, mtx, csv, loom, cloupe, rds, or built-in datasets.
- How to start: account/login, installation, dataset search, upload, or example data access.
- Available analysis features: QC, normalization, PCA/UMAP, clustering, marker genes, differential expression, annotation, spatial visualization.
- Visualization features: UMAP, gene expression plots, tissue/spatial view, metadata filtering, cluster exploration.
- Export options: figures, tables, processed data, reports, or session sharing.
- Strengths and limitations: what is easy, what is missing, and what still needs Python/R.

Practical Test

- Choose one real or example dataset inside the platform, or upload a small public dataset if the tool allows it.
- Show the main steps from opening the platform to loading or selecting the dataset.
- Create at least one visualization, such as UMAP, clusters, marker gene expression, or spatial tissue view.
- Take screenshots of the workflow and important outputs.

Slides to Prepare

- Platform introduction and website link.
- Supported data types and file formats.
- How to access or load data.
- Main analysis and visualization features.
- Screenshots from your practical test.
- Strengths and limitations.
- Final recommendation: who should use this platform and why?

Final Question to Answer

Can this no-code platform replace Scanpy, Squidpy, or Seurat, or is it mainly useful for data exploration and early-stage analysis?

Note

For Nygen / ScarfWeb, first check whether a free account or demo access is available. If access is limited, document the access process and evaluate the platform based on available screenshots, documentation, or demo materials.